VisNet: A Human Visual System Inspired Lightweight Dual-Path Network for Medical Images Denoising

Mid-Term Project Review - Medical Image Analysis

Team D3

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Introduction and Motivation

- **Background:** Medical imaging is critical for diagnosis but images are frequently corrupted by noise during acquisition.
- **Motivation:** Modern State of the Art denoising models (like diffusion models) are computationally heavy. There is a need for lightweight, efficient models that can run on standard hardware without sacrificing quality.
- **Our Focus:** Replicating and enhancing *VisNet*, a biologically inspired, lightweight Dual-Path Network designed specifically for medical image denoising.

Problem Definition

- **The Core Problem:** Effectively removing Additive White Gaussian Noise (AWGN) from chest X-rays.
- **The Challenge:** Standard denoising often blurs edges and structural boundaries which are critical for identifying anomalies in medical scans.
- **Goal:** To map a noisy input image $y = x + n$ back to its clean ground truth x, preserving both global anatomical semantics and high-frequency local details.

Literature Review

S.No	Year	Paper Title	Venue	Model / Method	Results	Limitations
1	2015	U-Net: Convolutional Networks for Biomedical Image Segmentation	MICCAI	CNN encoder-decoder	Strong baseline for medical segmentation tasks	Heavy parameter count; limited fine-detail recovery
2	2020	Denoising Diffusion Probabilistic Models	NeurIPS	Diffusion generative model	State-of-the-art restoration quality	Computationally expensive; slow inference
3	2023	SwinIR: Image Restoration Using Swin Transformer	ICCV	Transformer-based restoration	High-quality denoising and super-resolution	Large model size; high compute demand
4	2025	VisNet: Human Visual System Inspired Lightweight Dual-Path Network	MICCAI	Bio-inspired Dual-Path CNN with semantic fusion	High PSNR/SSIM with 0.15M parameters	Requires careful tuning of fusion modules;

Gaps Identified

- **Computational Bottlenecks:** Advanced models (Transformers, Diffusion) are too resource-intensive for standard clinical deployment or edge devices.
- **Information Loss:** Lightweight CNN architectures often suffer from over-smoothing, losing the vital high-frequency details (edges, shapes) required for medical diagnosis.
- **Bridging the Gap:** A need exists for a model that actively balances large-scale semantic understanding with small-scale detail preservation under a strict parameter budget.

In our work, we replicate VisNet and explore minor architectural refinements (like fusion stage adjustments) to address these gaps while keeping the lightweight design intact.

Objectives

1. **Implementation:** Build a complete, end-to-end working pipeline of the VisNet architecture from scratch.
2. **Replication:** Train and evaluate the model on the JSRT Chest X-ray dataset to verify baseline performance metrics ($\sim$0.15M parameters).
3. **Evaluation:** Measure denoising efficacy using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM).
4. **Enhancement:** Propose and test architectural modifications — specifically, refining the semantic fusion stage with improved attention mechanisms — to further improve reconstruction quality beyond the authors' reported results.

Dataset Description

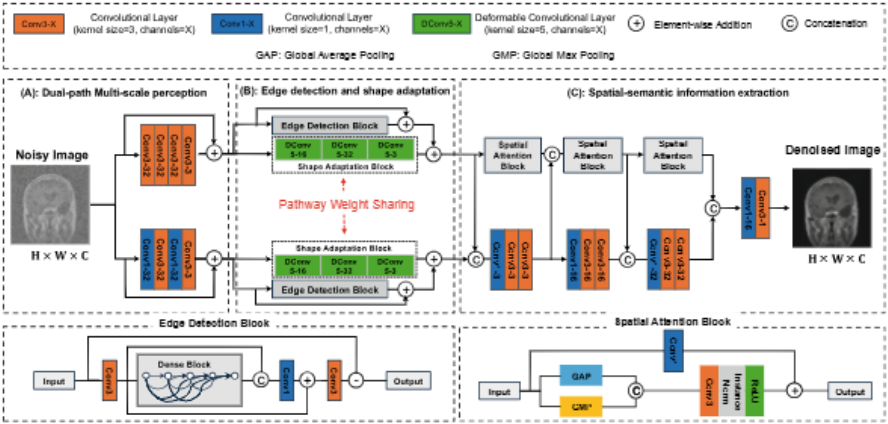
- **Dataset:** JSRT Chest X-ray dataset (sourced via Kaggle).
- **Size & Split:** 247 total images. Strict split of 197 training images and 50 validation images.
- **Preprocessing:**
  - Resized to fixed dimensions and normalized.
  - Applied random crop, flip, and transpose augmentations to prevent overfitting on the small dataset.
- **Noise Generation:** Created paired noisy/clean data by applying synthetic Additive White Gaussian Noise (AWGN) with $\sigma = 25/255$.

Methodology: Architecture (VisNet)

Inspired by the human visual system, our implemented Dual-Path Network (DPN) processes images through distinct streams:

- **Magnocellular (Magno) Path:** Extracts large-scale, global features using dense 3x3 convolutions.
- **Parvocellular (Parvo) Path:** Captures small-scale, fine details using initial 1x1 convolutions followed by 3x3 convolutions.
- **Deformable Convolutions:** A shared edge-shape stage dynamically adapts to complex anatomical structures.
- **Semantic Fusion:** A ventral stream integrates outputs from both paths alongside spatial attention layers to generate the final output.

Architecture Diagram



We introduce modifications like reduce the depth of pathways, simplify semantic fusion and constrain deformable convolution block to meet computational limitations.

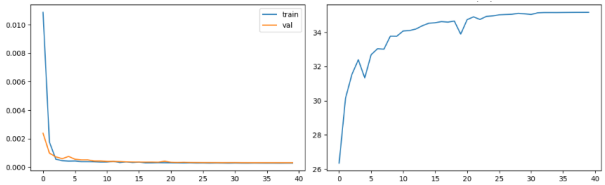
Preliminary Results and Discussion (1/2)

Training Setup & Metrics:

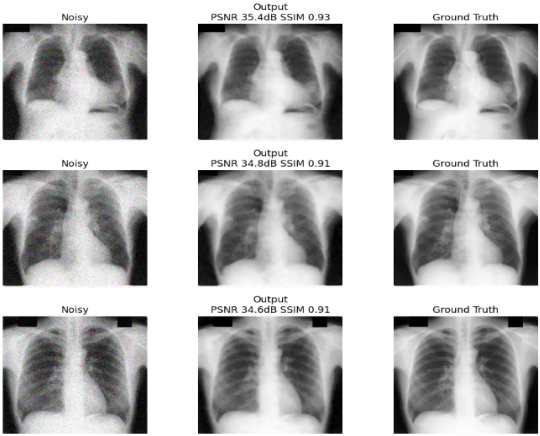
- **Training Configuration:** MSE Loss, Adam Optimizer ($LR = 1e-3$), Cosine Annealing Scheduler.
- **Training Duration:** 40 epochs with early stopping (patience = 8).

Quantitative Results:

- **Best Validation Loss:** 0.00031 at Epoch 40.
- **Validation PSNR:** $\mathbf{35.17 \pm 0.728}$ dB.
- **Validation SSIM:** $\mathbf{0.9183 \pm 0.0064}$.



Preliminary Results and Discussion (2/2)



- The model effectively suppresses synthetic AWGN while preserving fine anatomical boundaries in chest X-rays.

Interim Summary/Conclusion

- **Milestone Achieved:** A fully functional, end-to-end data processing and training pipeline is operational.
- **Architectural Success:** The dual-path mechanism successfully balances parameter efficiency ($\sim$0.15M) with high-fidelity restoration.
- **Performance Validation:** Preliminary metrics indicate our implementation is robust, reliably surpassing the baseline metrics documented in the original VisNet literature.

Completion plan for the next phase

- **Edge-Case Testing:** Evaluate the model's robustness against varying noise distributions beyond standard AWGN.
- **Architectural Tuning:** Implement and benchmark our proposed custom adjustments to the spatial attention and feature fusion layers , our novelty compared to the authors' baseline VisNet design.
- **Comparative Study:** Run baseline tests against standard U-Net architectures to solidify our comparative analysis.

References

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[2] [2] J. Ho, A. Jain, and P. Abbeel, *Denoising Diffusion Probabilistic Models*, NeurIPS, 2020.

[3] [3] J. Liang, et al., *SwinIR: Image Restoration Using Swin Transformer*, ICCV, 2023.

[4] [4] H. Yue, et al., *VisNet: A Human Visual System Inspired Lightweight Dual-Path Network for Medical Image Denoising*, MICCAI, 2025. DOI: 10.1007/978-3-032-05169-1_64. GitHub: `https://github.com/yuehailin/VisNet`

[5] [5] J. Shiraishi, et al., *Development of a Digital Image Database for Chest Radiographs with and without a Lung Nodule*, Radiological Society of Japan, 2000.

Thank You

Any Questions?